

Course Outline

MAD 411 Fundamentals of Android Development 1

Course ID: 030554**Academic Year:** 2024/25**Course Description:**

This course is designed to introduce students to both beginner application development for the Android operating system (OS), and the Kotlin programming language. Students will begin to learn the syntax of the language and how control, selection, and repetition structures are used in Kotlin. Students will study the history and components of the Android operating system and its applications. Students will also study the life cycle of Android applications and their use in diagnostic and debugging of applications. Students will learn to become proficient with integrated development environments (IDE's) and how they can be used to create and manage applications utilizing different layout designs

Pre-Requisites: MAD 100**Category:****Co-Requisites:** None**Course Credits:** 4.00**Special Conditions:** None**Academic Level:**

Instructional Hours:	Classroom Instruction	30
	Laboratory/Workshops	30
	Other	0
	Total Hours	60

Academic Department:

Windsor:	Zekelman School of Information Technology
Chair:	Ruth Susannah
Chatham:	
Chair:	

Revised By: Mathew Haug**Last Revision:** 06/01/2024

Required Tools, Equipment, and Learning Resources:

Mobile App Development with Android and Kotlin
zyBooks

Essential Employability Skills (EES):

Description	Teach	Assess
1) Communication: Communicate clearly, concisely and correctly in the written, spoken, and visual form that fulfills the purpose and meets the needs of the audience.		
2) Communication: Respond to written, spoken, or visual messages in a manner that ensures effective communication		
3) Numeracy: Execute mathematical operations accurately		
4) Critical Thinking: Apply a systematic approach to solve problems	✓	✓
5) Critical Thinking: Use a variety of thinking skills to anticipate and solve problems	✓	✓
6) Information Management: Locate, select, organize, and document information using appropriate technology and information systems	✓	✓
7) Information Management: Analyze, evaluate and apply relevant information from a variety of sources	✓	✓
8) Interpersonal: Show respect for the diverse opinions, values, belief systems, and contributions of others	✓	✓
9) Interpersonal: Interact with others in groups or teams in ways that contribute to effective working relationships and the achievement of goals		
10) Personal: Manage the use of time and other resources to complete projects	✓	✓
11) Personal: Take responsibility for one's own actions, decision and consequences	✓	✓

Course Learning Outcomes (CLO):

Upon successful completion of this course, the student will be able to:
(EKS = Embedded Knowledge and Skills)

- Construct a command line application with Kotlin (CLO #1)
EKS:
 - Employ the “while”, “do while” and “for” loops in Kotlin to allow for repetition of code
 - Employ the “when”, “if”, “if else”, “else if” selection structures to allow for decisions
 - Employ the “||”, “&&”, “==”, “!=”, “>”, “<”, “>=”, “<=”, “!” relational and logical operators to optimize selection

- Define and instantiate a class
- Implement Unit Testing with Kotlin
- Incorporate industry best practices into the creation of simple android applications with Kotlin (CLO #2)
EKS:
 - List the components of the Android Software Stack
 - Identify OS distribution and market share
 - Utilize OS distribution and market share information to build relevant applications for the appropriate audience
 - Construct an application that utilizes fragments and activities
 - Implement best practice when designing for accessibility
 - Utilize the Integrated Development Environment (IDE) debugger to troubleshoot code
 - Locate required information and resources from Android Developer Website
 - Utilize commenting for documentation where appropriate
- Construct an application that utilizes different elements to allow for user interaction (CLO #3)
EKS:
 - Employ the TextView xml element to display information to the screen
 - Employ the Button xml element to allow for basic user interaction
 - Employ the Image to display images to the screen
 - Employ the SeekBar to allow the user to select ranges
 - Employ the ViewPager to allow for pagination of content
 - Employ the EditText to allow user input
- Construct a simple application that utilizes simple layouts to provide a user interface (CLO #4)
EKS:
 - Employ LinearLayout to provide a horizontal or vertical layout
 - Employ the GridLayout to provide a structured grid layout
 - Employ a ConstraintLayout to allow for flexible and adaptable layouts
 - Employ a FrameLayout to allow for full screen layouts
 - Use multiple layouts that will interact with activities, fragments and destinations as needed
 - Employ various XML elements and attributes that are needed to build a desired layout

Teaching/Learning Activities:

Activities may include, but are not limited to:

Labs/Computer Labs

Lecture

Assessment:**Standard/Traditional Delivery**

Overall Assessment Notes: The assessment listed in this outline represents the planned assessment method for this course. Unanticipated conditions during the delivery of the course may necessitate changes to the planned assessment. Students will receive reasonable advance notice should any changes be necessary.

- Assignments 45.00%
 - Frequency:* 6-10
 - Description:* Assignments require problem analysis and the development of programming solutions. Some assignments may require individuals to work in groups. A portion of assignments may be thematically grouped as a project, although each assignment will be graded individually
 - Outcomes Assessed:* 1, 2, 3, 4
 - EES Assessed:* 4, 5, 6, 7, 8, 10, 11
- Final Exam 25.00%
 - Frequency:* 1
 - Description:* Exams may consist of both theory and applied programming techniques.
The theory portion may contain multiple-choice, fill in the blank, true/false, matching definition and short answer style questions. The applied portion requires students to solve short case problems by developing algorithms and writing the required program code to solve the problem.
 - Outcomes Assessed:* 1, 2, 3, 4
 - EES Assessed:* 4, 5, 6, 7, 8, 10, 11
- Midterm Test 20.00%
 - Frequency:* 1
 - Description:* Exams may consist of both theory and applied programming techniques.
The theory portion may contain multiple-choice, fill in the blank, true/false, matching definition and short answer style questions. The applied portion requires students to solve short case problems by developing algorithms and writing the required program code to solve the problem.
 - Outcomes Assessed:* 1, 2, 3
 - EES Assessed:* 4, 5, 6, 7, 8, 10, 11
- Topic Quizzes 10.00%
 - Frequency:* 6-10
 - Description:* Self-test online quizzes used to progressively evaluate the level of understanding of each topic covered in the course.
 - Outcomes Assessed:* 1, 2, 3, 4

EES Assessed: 4, 5, 6, 7, 10

100%

Note: The assessment listed in this outline represents the planned assessment method for this course. Unanticipated conditions during the delivery of the course may necessitate changes to the planned assessment. Students will receive reasonable advance notice should any changes be necessary.

Grading:

A = 80 - 100%
B = 70 - 79%
C = 60 - 69%
D = 50 - 59%
F = Less than 50%

Course Content:

See EKS statements

"Academic misconduct, including cheating of any form, will not be tolerated. Consequences may include, but are not limited to, a warning, a grade of "0" on the assignment/test/examination, or a failing grade in the course."

(Code of Students Rights and Responsibilities: Section 7.1.6)

All students and employees of this College have a right to study and work in an environment that is free from harassment and discrimination.

Accommodation Statement

The College will provide supports and services to all students with disabilities, both temporary and permanent, with valid supporting documentation. Interim accommodation requests will be received in good faith and can be provided pending receipt of medical documentation. Retroactive accommodations will be considered based on the unique circumstances of the individual matter. The College will give all Human Rights Code-related requests for accommodation meaningful consideration.

Procedure: The student is responsible to meet with a counsellor in Accessibility Services to discuss their functional limitations and accommodation needs and provide Accessibility Services with supporting documentation. Students are not required under the Ontario Human Rights Code to disclose their disability diagnosis (with the exception of Learning Disabilities) to receive accessibility supports and services and/or academic accommodations. Students are encouraged to meet with a counsellor prior to the start of a semester to provide information and arrange accommodations.